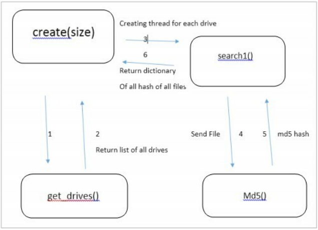

Figure 1: Code flow to create hashes of all files

The *md5()* function calculates the MD5 hash of the file.

```
def md5(fname,size=4096):
    hash_md5 = hashlib.md5()
    with open(fname, "rb") as f:
        for chunk in iter(lambda: f.read(size), b""):
            hash_md5.update(chunk)
    return hash_md5.hexdigest()
```

The *all_duplicate()* function in the following code is used to print all duplicate files in the drive. It gives the output to a file named *duplicate.txt* in the current running folder.

```
def all_duplicate(file_dict, path=""):

            file_txt = open('duplicate.txt', 'w')
            all_file_list = [v for k,v in file_dict.items()]
            for each in all_file_list:
                        if len(each)>2:
                       file_txt.write("------------------\n")
                                  for i in each:
                                              str1 = i+"\n"
                                              file_txt.
write(str1)
            file_txt.close()
```

The *get_drives()* function shown below returns the list of all drives. If you insert a pen drive or external drive while running the program, it will also be listed.

```
def get_drives():
     response = os.popen("wmic logicaldisk get caption")
            list1 = []
            total_file = []
            t1= datetime.now()
            for line in response.readlines():
                        line = line.strip("\n")
                        line = line.strip("\r")
                        line = line.strip(" ")
                        if (line == "Caption" or
line == ""):
                                    continue
                        list1.append(line)
            return list1
```

The *search1()* function shown in the following code gets the drive's name from the above function and accesses all files. It then sends the file with the full path to the *md5()* function to get the MD5 hash. Finally, it builds the global default dictionary *dict1*.

```
def search1(drive,size):
     for root, dir, files in os.walk(drive, topdown = True):
         try:
                            for file in files:
                                      try:
                             if os.access(root, os.X_OK):
           orig = file
           file = root+"/"+file
           if  os.access(file, os.F_OK):
              if os.access(file, os.R_OK):
                          s1=md5(file,size)
                          dict1[s1].append(file)
           except Exception as e :
                      pass
       except Exception as e :
           pass
```

The *create()* function is what starts to create the hashes of all files. It creates the threads for each drive and calls the *search1()* function. After the termination of each thread, the *create()* function dumps the default dictionary *dict1* to the pickle file named *mohit.dup1*, as shown below:

```
def create(size):
            t1= datetime.now()
            list2 = []   # empty list is created
            list1 = get_drives()
            print "Drives are \n"
            for d in list1:
                        print d," " ,
            print "\nCreating Index..."
            for each in list1:
                        process1 = Thread(target=search1,
args=(each,size))
                        process1.start()
                        list2.append(process1)

            for t in list2:
                        t.join() # Terminate the threads
```